BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Sound and Compression

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does a bitmap represent pixel colours?

2. Why can heavy virtual-memory use slow a program?
   A) RAM becomes permanent storage
   B) The CPU loses every core
   C) All instructions are deleted
   D) Secondary storage is slower

3. "A parameter is the final printed output of every program." Correct this. Explain why.

4. A microphone records digital sound. "Sampling stores every instant of the sound wave." Correct this. How does it affect recorded detail?

Today's objectives

Recall the essential ideas of sound and compression. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Sampling
Measuring an analogue signal at regular time intervals.

Sample rate
Number of samples recorded each second.

Bit depth
Number of bits per sample.

Lossless compression
Reduces size without losing original information.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Analogue to digital

An analogue sound wave is sampled at regular time intervals, and each sample is quantised to an available binary level.

Sample rate

Sample rate counts samples taken per second. More samples can capture changes in the wave more accurately.

Bit depth and size

Bit depth controls possible amplitude levels. Uncompressed mono size in bits = sample rate x duration x bit depth.

Compression

Lossless methods reconstruct original data exactly; lossy methods discard some information to achieve smaller files.

I Do - Worked model

Calculate the uncompressed size of a 10-second mono recording sampled at 8000 Hz with 8-bit depth. [4 marks]

- Identify three factors
- 8000 samples/second, 10 seconds, 8 bits/sample.
- Total samples = 8000 x 10 = 80,000.
- Size = 80,000 x 8 = 640,000 bits = 80,000 bytes. [4]

We Do - Guided practice

1. Calculate the uncompressed size of 5 seconds of mono audio at 16,000 samples/s and 8 bits/sample. [4]

2. Explain the effect of increasing sampling rate. [4]

3. Compare lossy and lossless compression for an audio archive. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the number of bits in a mono uncompressed recording.

```
rate = 800
seconds = 2
bitDepth = 8
bits = ...
print(bits)
```

You Do - Independent exam questions

Q2. A mono recording lasts 4 seconds, with 4000 samples/second and 8 bits/sample. Calculate the uncompressed size in bytes. [4]

Q3. Explain one effect of increasing bit depth while sample rate and duration stay the same. [3]

Q4. Explain why lossless compression is more suitable than lossy compression for an original master recording. [3]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Each pixel has a colour code stored in binary.

2. D - Secondary storage is slower

3. A named input received by a subprogram when called with an argument.

4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do - Explain the model

Identify three factors

8000 samples/second, 10 seconds, 8 bits/sample.

Total samples = 8000 x 10 = 80,000.

Size = 80,000 x 8 = 640,000 bits = 80,000 bytes. [4]

We Do - Model answers

1. 16,000 x 5=80,000 samples [1]; x 8=640,000 bits [2]; / 8=80,000 bytes [1].

2. More samples each second [1] can capture the original sound wave more accurately [1], but increases uncompressed data stored per second [1] and therefore file size [1].

3. Lossless preserves all original information [1] and can restore identical sound data [1]; lossy removes some information [1] for potentially greater size reduction but cannot restore exact original [1].

Code First - Answer

```
rate = 800
seconds = 2
bitDepth = 8
bits = rate * seconds * bitDepth
print(bits)
```

Expected output: 12800

You Do - Self-mark

Q2: 4000 x 4=16,000 samples [1]; x 8=128,000 bits [2]; / 8=16,000 bytes [1].

Q3: More amplitude levels [1] can improve quantisation precision [1], while increasing uncompressed file size [1].

Q4: Lossless preserves all original data [1], so the original can be reconstructed exactly [1], avoiding irreversible quality loss [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.